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END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech CSE   Semester: V

Name of the Paper: **Data Science : Visualization with Tableau Specialization**

Paper Code: **TCS 549**

Time: 3 Hours   Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub-questions among a, b & c in each main question

(iii) The total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	A city's food safety department has noticed a sharp increase in food poisoning cases across different restaurants, raising concerns about possible contamination points. As a data analyst, you are provided with records of reported cases, restaurant locations, food supply chain routes, customer demographics, and recent temperature data. Explain how data visualization tools can be used to analyze this data, identify patterns or hotspots, and assist health officials in taking timely action to prevent future outbreaks.	CO 1,3
(b)	See below	CO 2

Country	Literacy Male (%)	Literacy Female (%)	Dropout Rate (%)	Urban Enrollment (%)	Rural Enrollment (%)	Education Budget (% GDP)
India	82	70	8	88	64	4.1
Nepal	76	62	10	85	59	3.8
Bangladesh	80	68	7	90	67	4.5
Sri Lanka	93	91	3	96	88	3.5
Pakistan	79	61	12	82	55	2.8

Based on the data provided for India, Nepal, Bangladesh, Sri Lanka, and Pakistan, perform the following tasks:

1. Identify the top three countries with the highest overall literacy rate (average of male and female literacy).
2. Find the country with the largest male–female literacy gap.
3. Rank the countries according to their dropout rate to determine where the problem is most severe.
4. Calculate the urban–rural enrollment gap (urban − rural) for each country.
5. Identify the country with the smallest gender literacy gap (best gender parity).

After completing these tasks, present your findings in the form of:

- A suitable **title for your story** based on the results.
- One **table or chart** (can be hand-drawn) summarizing key data.
- **2–3 bullet points** highlighting major insights.
- One **policy suggestion or awareness idea** to improve education equity in the region.

(c) Human perception influences how effectively visuals communicate insights. As a designer, explain how principles of Gestalt perception, color theory, and preattentive attributes enhance the legibility and impact of visualizations. Include examples of poor perceptual design and their corrections. — CO 3

Q2 (20 marks)

(a) You are designing a public health dashboard on malnutrition. Explain how an editorial focus refines the visualization's narrative, ensuring clarity and emotional engagement. — CO 2

(b) Before visualizing education statistics, describe the process of data preparation and familiarization including formatting, cleaning, and deriving key ratios. Why is this process essential for meaningful storytelling? — CO 1

(c) Explain how visual analysis helps uncover hidden stories in multi-dimensional datasets for example, detecting gender bias in hiring trends or voter turnout patterns. Suggest visualization techniques suited for exploratory analysis. — CO 2.4

Q3 (20 marks)

(a) You are provided with the following dataset representing stock values (in ₹) over three days.

Dataset: Stock values (₹).

Day	Open	High	Low	Close
Mon	100	110	95	105
Tue	105	115	100	110
Wed	110	120	105	118

Using the above dataset, draw a candlestick chart to represent the daily stock price movements, showing the Open, High, Low, and Close values for each day. Clearly label the axes and days. After plotting, determine on which day the stock closed at the highest value and explain how the chart helps in identifying this trend. — CO 2,6

(b) You are provided with the following dataset representing scores of different skills on a scale of 1 to 10.
Dataset:

Skill	Score
Communication	7
Coding	9
Design	6
Management	8

Using the above dataset, draw a Radial (Spider/Radar) Chart to represent the performance of each skill. Clearly mark each axis with the skill name and plot the corresponding score values After completing the chart, identify which skill has the highest score and explain how the chart visually indicates this. — CO 2,6

(c) A logistics company wants to analyze supplier relationships and product movement between cities. Suggest suitable visualizations (such as scatter plot, network graph, or Sankey diagram). Explain how each visualization helps in identifying bottlenecks or inefficiencies. — CO 2

Q4 (20 marks)

(a) Discuss the key design considerations for building a collaborative visual analytics system that supports both real-time and asynchronous teamwork. — CO 5

(b)	A design team is developing a collaborative crime-analysis visualization tool. Discuss key design considerations for effective teamwork.	CO 3,5
(c)	List any two software tools used for data visualization and explain how they help in presenting data clearly.	CO 2
Q5	(20 marks)	
(a)	You have datasets stored in multiple formats: CSV (demographics), Excel (budget), and SQL (employment). Explain the steps to connect Tableau to multiple data sources, clean data, and create a unified visualization.	CO 1,6
(b)	You are a data analyst combining SQL, Excel, and AWS data in Tableau to create a real-time sales dashboard. Explain how Tableau's Data Layer, Processing Layer, and Client Layer interact in this process.	CO 6
(c)	A healthcare company wants to visualize patient data securely using Tableau connected to cloud and hospital databases. Describe how the Data Server, Repository, and Tableau Online support secure data handling and sharing.	CO 6